Week 09 – Assignment

Scenario: Product Review Analysis

An online marketplace wants to analyze the words used in customer reviews. The program tokenizes the review text, creates a frequency distribution using FreqDist, and displays only the words that occur more than once. It also reports the total number of unique words.

```
1. from nltk.tokenize import word_tokenize
2. from nltk import FreqDist
3.
4. reviews = """
5. Fast delivery and good packaging.
6. Good quality and fast service.
7. Delivery was fast and packaging was good.
8. """
9.
10. tokens = word_tokenize(reviews.lower())
11. freq = FreqDist(tokens)
12.
13. print(len(tokens))
14. print(len(freq))
15.
16. for word in sorted(freq):
17.     if freq[word] > 1:
18.         print(word, freq[word])
```

Q1. Which statement converts the review text into a list of tokens?

A. Line 10

B. Line 11

C. Line 13

D. Line 16

Answer: A

Justification: word_tokenize() splits the given text into individual tokens and stores them in the list tokens.

Q2. What does the expression len(freq) in Line 14 return?

A. Total number of characters in the review

B. Number of unique tokens in the review

C. Total number of repeated words

D. Number of lines in the review

Answer: B

Justification: A FreqDist object stores one entry for each distinct token, so len(freq) returns the number of unique tokens.

Q3. Which line creates the frequency distribution of the tokens?

A. Line 10

B. Line 16

C. Line 11

D. Line 13

Answer: C

Justification: FreqDist(tokens) counts the occurrences of every token and stores the resulting frequency distribution.

Q4. Which part of the program prints only the words that occur more than once?

A. Line 10

B. Line 13

C. Line 14

D. Lines 16–18

Answer: D

Justification: The loop checks each token's frequency and prints only those whose count is greater than 1.

Q5. (MSQ) Which of the following statements are correct?

A. len(tokens) returns the number of unique words in the review.

B. word_tokenize() automatically counts the frequency of each token.

C. FreqDist stores each token as a key and its frequency as the corresponding value.

D. len(freq) gives the number of distinct tokens present in the text.

Answer: C and D

Justification: FreqDist behaves like a dictionary with tokens as keys and counts as values, and len(freq) returns the number of distinct keys stored in the distribution.

Scenario: Library Book Recommendation Network

A digital library recommends books based on their similarity. Each book is represented as a node, and an edge between two books indicates that they are frequently borrowed together. The librarian uses NetworkX to model and examine this recommendation network.

```
import networkx as nx

G = nx.Graph()

G.add_nodes_from(["Python", "AI", "ML", "Data", "Cloud"])

G.add_edges_from([
    ("Python", "AI"),
    ("Python", "Data"),
    ("AI", "ML"),
    ("ML", "Data"),
    ("Cloud", "Data")
])

print(len(G.nodes))
print(len(G.edges))

for book in G.nodes:
    print(book, list(G.neighbors(book)))
```

Q1. In this graph, what does a node represent?

A. A book

B. A recommendation between two books

C. The total number of books

D. A borrowing record

Answer: A

Justification: Each node represents one book in the recommendation network.

Q2. What does an edge between two books indicate?

A. The books have the same author.

B. The books belong to different categories.

C. The books are borrowed together and are recommended to each other.

D. The books have the same publication year.

Answer: C

Justification: Each edge represents a recommendation relationship between two books.

Q3. What does len(G.nodes) represent?

A. Number of recommendations

B. Number of books in the network

C. Number of neighboring books

D. Number of connected components

Answer: B

Justification: G.nodes contains all the books (nodes) present in the graph.

Q4. Why is G.neighbors(book) used in the program?

A. To remove duplicate books

B. To sort the books alphabetically

C. To calculate the shortest path

D. To find books directly connected to a given book

Answer: D

Justification: neighbors() returns all nodes that share an edge with the given node.

Q5. (MSQ) Which of the following statements are correct?

A. Every node must be connected to every other node.

B. An edge can exist without any nodes.

C. len(G.edges) returns the total number of recommendation links in the graph.

D. A graph can be used to model relationships between connected objects.

Answer: C and D

Justification: Edges represent relationships between nodes, and graphs are commonly used to model such connections.

Scenario: Six Degrees of Separation

A social networking platform represents its users as a graph. Each person is a node, and a friendship between two people is represented by an edge. According to the "Six Degrees of Separation" concept, any two people in a social network can usually be connected through a small chain of mutual friends. The platform analyzes these connections to recommend new friends and understand how closely users are connected.

Q1. In the social network described above, what does an edge represent?

A. A user profile

B. A friendship between two users

C. The total number of users

D. A friend recommendation

Answer: B

Justification: An edge represents a direct friendship connecting two users.

Q2. Alice is friends with Bob, and Bob is friends with Carol. Alice and Carol are not directly connected. What exists between Alice and Carol?

A. A path through Bob

B. A duplicate edge

C. A disconnected graph

D. A self-loop

Answer: A

Justification: A path connects two nodes through one or more intermediate nodes.

Q3. Why is the shortest path between two users often important in a social network?

A. It increases the total number of users.

B. It removes duplicate friendships.

C. It identifies the minimum number of friendship links connecting two users.

D. It converts the graph into a directed graph.

Answer: C

Justification: The shortest path represents the fewest intermediate connections between two users.

Q4. When is a social network considered to be connected?

A. Every user has exactly one friend.

B. Every user is directly connected to every other user.

C. The graph has no edges.

D. A path exists between every pair of users.

Answer: D

Justification: A connected graph allows every node to be reached from every other node through at least one path.

Q5. (MSQ) Which of the following statements are correct?

A. Every pair of users must have a direct friendship for a path to exist.

B. A graph cannot have more than one path between two users.

C. A node represents an individual user in the social network.

D. The Six Degrees of Separation concept is based on finding paths between people in a network.

Answer: C and D

Justification: In a social graph, nodes represent users, and the Six Degrees concept relies on paths that connect people through mutual friends.

Scenario: Quality Inspection of Product Images

An online retail company inspects product images before publishing them on its website. The system opens an image, checks the color of a specific pixel, marks that pixel with a new color for verification, creates a small thumbnail image, and saves the modified image.

```
1. from PIL import Image
2.
3. img = Image.open("product.jpg")
4.
5. print(img.size)
6. print(img.getpixel((20, 20)))
7.
8. img.putpixel((20, 20), (255, 0, 0))
9.
10. thumb = Image.new("RGB", (100, 100), "white")
11. thumb.save("thumbnail.jpg")
12.
13. img.save("verified_product.jpg")
```

Q1. Why is getpixel() used in this program?

A. To resize the image

B. To change the image dimensions

C. To retrieve the color value of a specific pixel

D. To save the image

Answer: C

Justification: getpixel() returns the color value stored at the specified pixel location.

Q2. What is the purpose of putpixel()?

A. To create a new image

B. To modify the color of a specific pixel

C. To compress the image

D. To display the image

Answer: B

Justification: putpixel() updates the color value of a specified pixel.

Q3. What is the purpose of Image.new()?

A. To open an existing image

B. To copy an existing image

C. To convert an image into grayscale

D. To create a new blank image

Answer: D

Justification: Image.new() creates a new image with the specified mode, size, and background color.

Q4. Why is the save() method used?

A. To store the processed image as a file

B. To display the image on the screen

C. To remove unwanted pixels

D. To count the number of pixels

Answer: A

Justification: The save() method writes the image object to a file.

Q5. (MSQ) Which of the following statements are correct?

A. getpixel() changes the color of a pixel.

B. Image.new() creates a new image with the specified mode and dimensions.

C. putpixel() automatically saves the modified image to disk.

D. Changes made using putpixel() are stored in memory until the image is saved.

Answer: B and D

Justification: Image.new() creates a new image object, and modifications made using putpixel() become permanent only after calling save().